

## 4.6.2 Electromagnetic waves

### 4.6.2.1 Types of electromagnetic waves

| Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Key opportunities for skills development |            |               |               |             |            |            |             |            |          |               |             |        |            |  |
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| <p>Electromagnetic waves are transverse waves that transfer energy from the source of the waves to an absorber.</p> <p>Electromagnetic waves form a continuous spectrum and all types of electromagnetic wave travel at the same velocity through a vacuum (space) or air.</p> <p>The waves that form the electromagnetic spectrum are grouped in terms of their wavelength and their frequency. Going from long to short wavelength (or from low to high frequency) the groups are: radio, microwave, infrared, visible light (red to violet), ultraviolet, X-rays and gamma rays.</p> <div><div>Long wavelength</div><div><table><tr><td>Radio waves</td><td>Microwaves</td><td>Infrared</td><td>Visible light</td><td>Ultraviolet</td><td>X-rays</td><td>Gamma rays</td></tr></table></div><div>Short wavelength</div></div> <div><div>Low frequency</div><div><table><tr><td>Radio waves</td><td>Microwaves</td><td>Infrared</td><td>Visible light</td><td>Ultraviolet</td><td>X-rays</td><td>Gamma rays</td></tr></table></div><div>High frequency</div></div> <p>Our eyes only detect visible light and so detect a limited range of electromagnetic waves.</p> <p>Students should be able to give examples that illustrate the transfer of energy by electromagnetic waves.</p> | Radio waves                              | Microwaves | Infrared      | Visible light | Ultraviolet | X-rays     | Gamma rays | Radio waves | Microwaves | Infrared | Visible light | Ultraviolet | X-rays | Gamma rays |  |
| Radio waves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Microwaves                               | Infrared   | Visible light | Ultraviolet   | X-rays      | Gamma rays |            |             |            |          |               |             |        |            |  |
| Radio waves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Microwaves                               | Infrared   | Visible light | Ultraviolet   | X-rays      | Gamma rays |            |             |            |          |               |             |        |            |  |

### 4.6.2.2 Properties of electromagnetic waves 1

| Content                                                                                                                                                                                                                                                                                                                                                                                                                 | Key opportunities for skills development |
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| <p>(HT only) Different substances may absorb, transmit, refract or reflect electromagnetic waves in ways that vary with wavelength.</p> <p>(HT only) Some effects, for example refraction, are due to the difference in velocity of the waves in different substances.</p> <p>Students should be able to construct ray diagrams to illustrate the refraction of a wave at the boundary between two different media.</p> |                                          |
| <p>(HT only) Students should be able to use wave front diagrams to explain refraction in terms of the change of speed that happens when a wave travels from one medium to a different medium.</p>                                                                                                                                                                                                                       | WS 1.2                                   |

**Required practical activity 10:** investigate how the amount of infrared radiation absorbed or radiated by a surface depends on the nature of that surface.

AT skills covered by this practical activity: AT 1 and 4.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in [Key opportunities for skills development](#) (page 98).

#### 4.6.2.3 Properties of electromagnetic waves 2

| Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Key opportunities for skills development |
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| <p>(HT only) Radio waves can be produced by oscillations in electrical circuits.</p> <p>(HT only) When radio waves are absorbed they may create an alternating current with the same frequency as the radio wave itself, so radio waves can themselves induce oscillations in an electrical circuit.</p> <p>Changes in atoms and the nuclei of atoms can result in electromagnetic waves being generated or absorbed over a wide frequency range. Gamma rays originate from changes in the nucleus of an atom.</p> |                                          |
| <p>Ultraviolet waves, X-rays and gamma rays can have hazardous effects on human body tissue. The effects depend on the type of radiation and the size of the dose. Radiation dose (in sieverts) is a measure of the risk of harm resulting from an exposure of the body to the radiation.</p>                                                                                                                                                                                                                      | WS 1.5                                   |
| <p>1000 millisieverts (mSv) = 1 sievert (Sv)</p> <p>Students will not be required to recall the unit of radiation dose.</p> <p>Students should be able to draw conclusions from given data about the risks and consequences of exposure to radiation.</p> <p>Ultraviolet waves can cause skin to age prematurely and increase the risk of skin cancer. X-rays and gamma rays are ionising radiation that can cause the mutation of genes and cancer.</p>                                                           | WS 1.5                                   |

#### 4.6.2.4 Uses and applications of electromagnetic waves

| Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Key opportunities for skills development |
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| <p>Electromagnetic waves have many practical applications. For example:</p> <ul style="list-style-type: none"> <li>• radio waves – television and radio</li> <li>• microwaves – satellite communications, cooking food</li> <li>• infrared – electrical heaters, cooking food, infrared cameras</li> <li>• visible light – fibre optic communications</li> <li>• ultraviolet – energy efficient lamps, sun tanning</li> <li>• X-rays and gamma rays – medical imaging and treatments.</li> </ul> |                                          |

| Content                                                                                                                                       | Key opportunities for skills development |
|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| (HT only) Students should be able to give brief explanations why each type of electromagnetic wave is suitable for the practical application. | (HT only) WS 1.4                         |

#### 4.6.2.5 Lenses (physics only)

| Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Key opportunities for skills development                                                                                                                                                                      |
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| <p>A lens forms an image by refracting light. In a convex lens, parallel rays of light are brought to a focus at the principal focus. The distance from the lens to the principal focus is called the focal length. Ray diagrams are used to show the formation of images by convex and concave lenses.</p> <p>The image produced by a convex lens can be either real or virtual. The image produced by a concave lens is always virtual.</p>                                                                                         |                                                                                                                                                                                                               |
| <p>Students should be able to construct ray diagrams to illustrate the similarities and differences between convex and concave lenses.</p> <p>The magnification produced by a lens can be calculated using the equation:</p>                                                                                                                                                                                                                                                                                                          | <p>MS 5a, 5c</p> <p>WS 1.2</p>                                                                                                                                                                                |
| <p><math display="block">\text{magnification} = \frac{\text{image height}}{\text{object height}}</math></p> <p>Magnification is a ratio and so has no units.</p> <p>Image height and object height should both be measured in either mm or cm.</p> <p>In ray diagrams a convex lens will be represented by:</p>  <p>A concave lens will be represented by:</p>  | <p>MS 3b, c</p> <p>Students should be able to apply this equation which is given on the Physics equation sheet.</p> <p>AT 4, 8</p> <p>Investigate the magnification produced by a range of convex lenses.</p> |

#### 4.6.2.6 Visible light (physics only)

| Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Key opportunities for skills development |
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| <p>Each colour within the visible light spectrum has its own narrow band of wavelength and frequency.</p> <p>Reflection from a smooth surface in a single direction is called specular reflection. Reflection from a rough surface causes scattering: this is called diffuse reflection.</p> <p>Colour filters work by absorbing certain wavelengths (and colour) and transmitting other wavelengths (and colour).</p> <p>The colour of an opaque object is determined by which wavelengths of light are more strongly reflected. Wavelengths that are not reflected are absorbed. If all wavelengths are reflected equally the object appears white. If all wavelengths are absorbed the objects appears black.</p> <p>Objects that transmit light are either transparent or translucent.</p> <p>Students should be able to explain:</p> <ul style="list-style-type: none"><li>• how the colour of an object is related to the differential absorption, transmission and reflection of different wavelengths of light by the object</li><li>• the effect of viewing objects through filters or the effect on light of passing through filters</li><li>• why an opaque object has a particular colour.</li></ul> |                                          |